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EDITORIAL

Sustainable production and consumption

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Abstract
The challenges of balancing environmental sustainability with economic and social growth are intensifying, with unsustainable consumption and production patterns posing significant risks on a global scale (i.e., resource scarcity, raw material consumption, waste production etc.). Projections indicate a sharp increase in municipal solid waste and associated costs by 2050 while combating all implications of this increase becomes more and more vital. To address these challenges, the European Green Deal and the Sustainable Development Goals (SDGs) of the United Nations, particularly SDG 12, emphasize the necessity of sustainable production and consumption of all products, raw materials, energy, water etc. This involves adopting a circular economy model, which transforms waste into resources, conserving natural resources, and enhancing environmental performance. Key strategies include broader implementation of R strategies (reduce, reuse, recycle, remanufacture, refurbish), producer responsibility, and advanced monitoring tools like Life Cycle Assessment (LCA) and Material Flow Analysis (MFA). Additionally, sustainable energy systems and innovative digital technologies (i.e. artificial intelligence, Internet of Things, blockchain etc.) and the holistic utilization of Industry 5.0 play a crucial role. Public education and awareness are essential for fostering sustainable behaviors and consumption patterns coupled with effective policy, governance and the building of trust between the public and decision makers. At the same time, collaborative efforts from governments, industries, and communities are imperative for transitioning to a sustainable and circular economy, ensuring a resilient and equitable future.

Keywords SDG12 · Sustainable development · Circular economy · Production and consumption

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Introduction

As the world struggles to maintain its environmental sustainability and balance it out with economic and social aspects around the globe, the dangers of unsustainable consumption and production become even more dominant. Municipal solid waste is expected to increase from 2.1 billion tons in 2023 to 3.8 billion tons by 2050. In 2020, the direct global cost of waste management was approximately 252 billion dollars. Without immediate and effective action on waste management, the annual global cost could nearly double to an astounding 640.3 billion dollars by 2050 (Fig. 1) (United Nations Environment Programme 2024).

In accordance with the goals of the European Green Deal for carbon neutrality until 2050 and the circular economy strategy, the EU continuously seeks to reduce waste and promote the reuse and recycling of materials. A baseline for these goals is the Sustainable Development Goals (SDGs) of the United Nations in 2015, is gaining momentum as a significant societal trend (Vázquez-Méndez et al. 2024). Environmental sustainability, a vital component of sustainable development, is explicitly emphasized in SDG 6 (Clean Water and Sanitation), SDG 13 (Climate Change), SDG 14 (Life below Water), and SDG 15 (Life on Land), as well as indirectly addressed in SDG 2 (Zero Hunger), SDG 7 (Affordable and Clean Energy), and SDG 11 (Sustainable cities and communities). Specifically, SDG 12 (sustainable consumption and production) constitutes one of the most important goals concerning waste management. However, SDG12 implementation require a variety of different disciplines, and sectors, including but not limited to sustainable cities, public behavior and awareness, environmental and economic sustainability, technology, governance, academia, research organizations and other (Hasnaoui et al. 2024).

Sustainable consumption and production

Sustainable production and consumption (SDG 12) can be defined as the production and use of products and services in a manner that is socially beneficial (i.e. health, education, equality), economically viable, and environmentally benign over their whole life cycle (Golia et al. 2023). In this regard, the circular economy model has been identified as a key approach to achieving sustainable development. Efforts are centered on implementing a circular economy that treats waste from one production process as a resource for another (closing the loop), thereby conserving natural resources (i.e. fossil fuels, raw materials, waste etc.) and enhancing environmental quality (Zhang and Nakatani 2024).

When it comes to circular economy however, a broader and more holistic understanding and application of the R strategies (i.e. reduce, reuse etc.) must be in place. Limiting the utilization of circular economy logic to 'reduce,

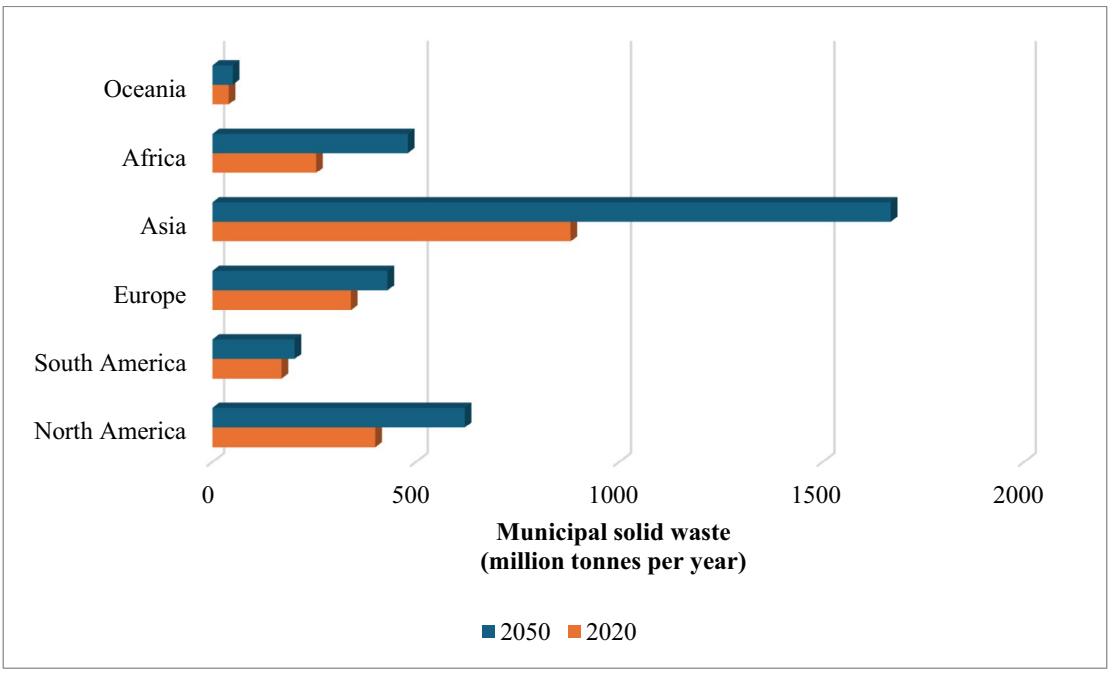


Fig. 1 Estimated projected municipal solid waste generation from 2020 to 2025 for the six continents (Antarctica was assumed to produce no waste) (Figure created by the authors) (Data from: United Nations Environment Programme 2024)

reuse and recycle' is no longer an option (Papamichael et al. 2024). Other R strategies including but not limited to remanufacture, refurbish, redesign, rethink etc. are significant to be applied throughout the supply chain of a product (i.e. fashion, food waste, agriculture etc.) to ensure circularity to all aspects of production from raw material acquisition to use and disposal of the product (Kouvousi et al. 2024; Papamichael et al. 2024). In this regard, producer responsibility is a key principle for SDG 12 as it emphasizes the role of producers in ensuring that their products are designed and manufactured in a way that minimizes environmental impacts and promotes sustainable consumption patterns. Producers must take responsibility for the environmental and social impacts of their products throughout their entire life cycle, from raw material extraction to end-of-life disposal or recycling (Economou et al. 2024).

But the 'how' and 'why' a producer should concern themselves with sustainable production and consumption does not lie only with EU regulations, strategies and action plans. The correct monitoring and reporting of environmental implications of industrial activities as well as altering a business-as-usual scenario to a more circular or sustainable one (i.e., from linear production to circular, industrial symbiosis etc.) can both create a better corporal social responsibility and economic viability (Papamichael et al. 2024). Monitoring, consumption and production patterns is one of the first steps towards sustainability in every setting (industrial, urban, household, etc.). Using Key Performance Indicators (KPIs) as well as Life Cycle Assessment (LCA) and Material Flow Analysis (MFA) amongst other key tools (i.e., environmental management systems like ISO etc.) must be used in order for every stakeholder to have the full picture of the environmental, social and economic implications of their consumption and production patterns (Papamichael et al. 2023). Monitoring tools are crucial for identifying hotspots of waste production in different supply chains (i.e. food waste, fashion waste, etc.) as well as tracking and optimizing processes (Economou et al. 2024). In particular, LCA is a standardized method that examines the environmental aspects and potential impacts of products or services throughout their entire lifecycle, from raw material acquisition to production, use, end-of-life treatment, recycling, and final disposal (i.e., cradle-to-grave). Over the past three decades, LCA has evolved significantly from basic energy analysis to encompass full life cycle impact assessment, life cycle costing, and social LCA, culminating in a more comprehensive life cycle sustainability analysis that expands the scope of traditional environmental evaluations (Economou et al. 2024).

Simultaneously, sustainable energy systems are essential for achieving sustainable consumption and production. The strong dependency of humanity on fossil fuels makes different energy recovery technologies (i.e. waste valorization) and renewable energy sources use (i.e., solar, wind, biomass, geothermal etc.) as well as other water collection methods (i.e., fogging systems) an increasingly interesting approach for the EU (Elshennawy et al. 2024; Tsangas et al. 2024). SDG 12 includes improving resource efficiency, reducing waste, and decreasing environmental degradation. The anticipated growth of the global renewable energy market, projected to exceed 2 trillion dollars by 2030, directly supports SDG 12 by promoting the use of cleaner energy sources. China's significant electricity consumption (8000 terawatt-hours per year), though the highest globally, highlights the importance of improving energy efficiency and reducing per capita energy use (Statista 2024).

Technology and innovation

When it comes to SDG 12, in the mists of the 4th industrial revolution and or our way to the industry 5.0, digitalization encompasses not only the conversion of information into digital formats but also the application of digital capabilities to analog aspects of life to enhance them (Lehner et al. 2024). Leveraging the benefits of digitalization and the transition to virtual worlds (i.e., metaverse), can catalyze the development of smart systems and interactions of them with sustainability goals and SDG 12 (Papamichael et al. 2023). By enabling virtual testing that mirrors physical concepts, the metaverse can reduce physical consumption, promote the sharing economy, and educate users on sustainable behaviors. This can include Geographic Information System (GIS), artificial intelligence (i.e., robotics in agriculture), blockchain (i.e. product passport), virtual reality, augmented reality (i.e., virtual try on clothing in fashion), internet of things (i.e., pollution sensors of heavy metals (i.e., Cadmium) in soil etc.), smart waste systems (i.e., Pay as You throw), waste management platforms, and serious games in the context of gamification (Golia et al. 2023; Papadimou et al. 2024).

Despite various benefits of digitalization (i.e. transparency, security, monitoring, reporting, time management etc.), urban settings lack assessment tools for stakeholders to explore and implement digitalization strategies (Salvati and Vardopoulos 2024). The integration of KPIs (i.e., waste metrics, waste generation, production rate etc.), quality protocols, and digitalization offers an informative and holistic simulation, enabling users to understand and evaluate the environmental impact of various activities in urban settings (Lehner et al. 2024). However, the combination of digitalization with gamification can be vital for the clear understanding of waste generation by various stakeholders (i.e. academia, industry, policy makers, etc.). Gamification involves applying game design principles outside of traditional gaming contexts to motivate and captivate individuals in specific tasks. Utilizing gamified educational tools which focus on waste management in urban environments,

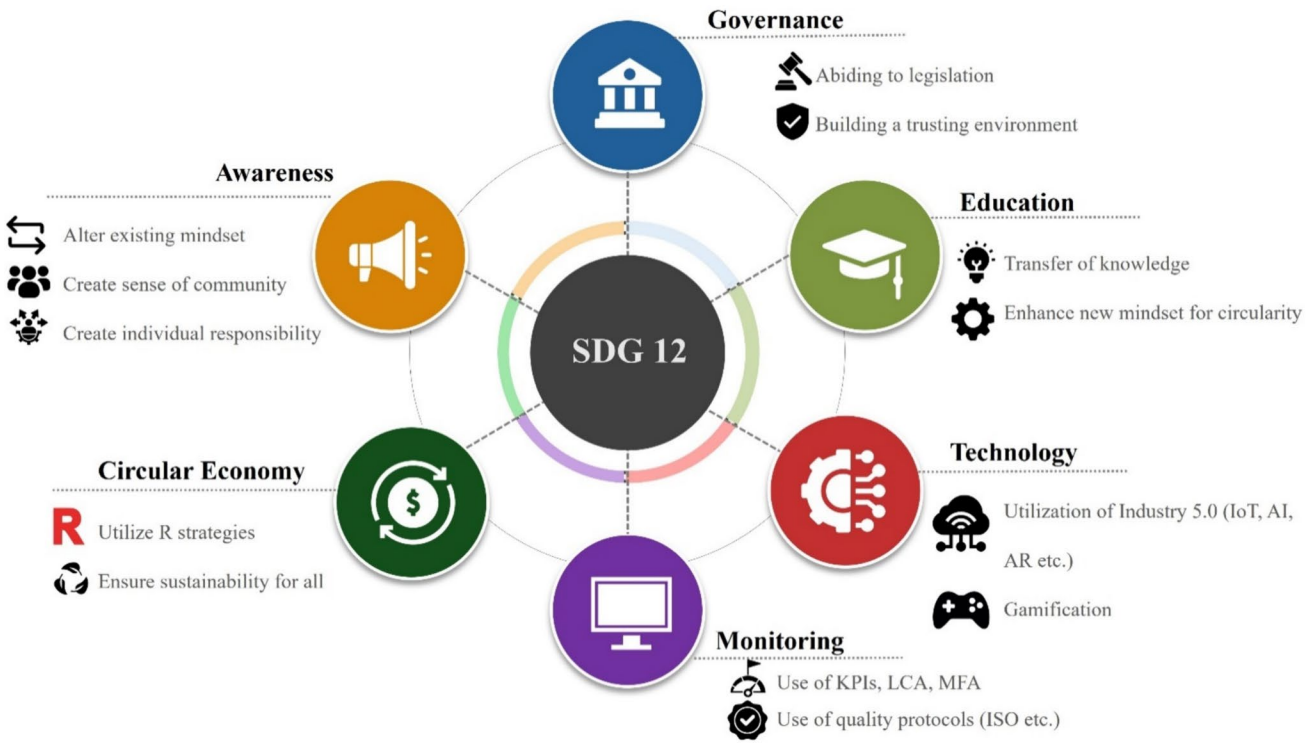


Fig. 2 Main areas of research regarding SDG 12 and responsible consumption and production (Figure created by the authors)

stakeholders can immerse in an interactive environment where they can actively engage, learn, and understand the implications of daily activities or urban planning approaches on waste management and sustainability (Papamichael et al. 2023).

Education and awareness

The importance of public education and awareness regarding consumption and production is in fact one of the most importance pillars towards sustainability (Emmanouil et al. 2024). Achieving SDG 12 requires a multifaceted approach involving consumer engagement, ethical investment, intra- and intergenerational equity, and sustainable urban development. The significance of educating the public about the environmental impacts (SDG 3) of their consumption choices and the importance of adopting sustainable practices. This awareness extends to the broader context of achieving SDG 12, which encompasses consumer engagement, ethical investment, intergenerational equity, and sustainable urban development (Zoubi et al. 2024).

However, enhancing consumer participation in sustainability requires a shift in behavior towards the use of renewable energy, organic products, and the principles of the R strategies of circular economy. Achieving SDG 12 necessitates a comprehensive understanding of the various social factors that influence consumption behaviors (Zoubi et al. 2024). Factors such as religion (i.e. dress code and fashion waste), wars (i.e. military waste, disruption of supply chain etc.), cultural norms (i.e. donating clothing), and societal values all play crucial roles in shaping consumption patterns. Shifting consumer behavior towards sustainability requires more than just individual choices; it requires systemic changes (Papamichael et al. 2024).

At the same time, monitoring social attitude and behavior towards circular and sustainable products is beneficial from all environmental, social, economic and marketing points of view. For instance, in the realm of circular fashion and the circular economy within the fashion industry, customer behavior emerges as a critical factor. Their Willingness to Pay (WTP) (willing to spend on acquiring a circular product-made from waste) vs their Ability to Pay (ATP) (financial capacity to purchase a circular product or service) can navigate the industry towards which and how many consumers are gravitating towards eco-friendlier and circular approaches when making purchases (Fraccascia et al. 2023).

Policy and governance

All aspects concerning SDG 12 need to be prioritized by decision makers and policy makers in order for anything to be taken at heart. One key aspect of this, involves the effective implementation of existing legislations amongst the other mentioned aspects towards achieving SDG 12

(Fig. 2). Governments must ensure rigorous enforcement of regulations related to waste management, resource conservation, and pollution control to drive tangible progress towards the SDGs. This requires collaboration between policymakers, regulatory bodies, industries, and civil society to monitor compliance, address loopholes, and adapt regulations to evolving environmental challenges (Zorpas et al. 2021).

At the same time, crucial aspect of policy and governance in advancing SDG 12 is the need to build trust between governments and their citizens. Trust is essential for fostering cooperation, transparency, and accountability in policymaking processes related to sustainable consumption and production. Governments must engage with citizens through participatory decision-making mechanisms, public consultations, and information dissemination to ensure inclusivity and legitimacy in policy formulation (Trantidis and Cowen 2024). By involving stakeholders in the policymaking process and addressing their concerns and preferences, governments can enhance public trust and confidence in sustainability initiatives. This, in turn, can facilitate the adoption of sustainable behaviors, encourage voluntary compliance with regulations, and promote collective action towards achieving SDG 12 targets (Zorpas et al. 2021).

Conclusion

The journey towards sustainable consumption and production is a multifaceted endeavor that requires collaborative efforts from various stakeholders, including governments, industries, communities, and individuals. By promoting responsible consumption and production patterns, reducing waste, and fostering the transition to a circular economy, strategies like the European Green Deal, Circular Economy Strategy and other aim to address pressing environmental challenges while promoting inclusive and sustainable growth. However, achieving sustainable consumption and production requires more than just policy and governance. It necessitates a shift in behavior, education, and awareness among consumers, businesses, and communities. Embracing green consumption practices, leveraging technology and innovation, and fostering a culture of sustainability are essential components of this transformation. As humanity navigates the challenges of the twenty-first century, the pursuit of sustainable consumption and production remains imperative for building a resilient and equitable future for all. By working together and embracing the principles of sustainability, we can pave the way towards a more prosperous and harmonious world for generations to come.

Declarations

Conflict of interest The authors declare that there is no conflict of interest.

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